YEAR 7 MATHS · CHAPTER 1 · MIND MAP

Computation with positive integers

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1 Ancient number systems

Roman numerals add up, but a smaller symbol before a larger one means subtract.

I = 1, V = 5, X = 10, L = 50, C = 100

XIV = 14 XL = 40 CXXIII = 123

2 Place value

Each digit's value depends on where it sits. A zero holds an empty place.

The 3 in 4325 is worth 300

$4325 = 4 \times 1000 + 3 \times 100 + 2 \times 10 + 5$

507: the zero holds the tens place

3 Adding and subtracting

Round to a friendly number, then adjust.

$47 + 28 = 47 + 30 - 2 = 75$

$63 - 18 = 63 - 20 + 2 = 45$

Order doesn't matter: $38 + 17 + 2 = 38 + 2 + 17 = 57$

4 Algorithms for + and −

Line up the place values, then carry or borrow.

$476 + 385 = 861$

$603 - 247 = 356$

Check: $356 + 247 = 603$ ✓

Watch out: line up the units, not the left-hand digits.

5 Multiplying small numbers

Split a number into tens and units. Order doesn't matter, so pair numbers that are easy.

$6 \times 23 = 6 \times 20 + 6 \times 3 = 138$

$4 \times 17 \times 25 = 100 \times 17 = 1700$

$5 \times 14 \times 2 = 10 \times 14 = 140$

6 Multiplying large numbers

Multiply by each digit, then add the parts.

$37 \times 24 = 37 \times 20 + 37 \times 4 = 740 + 148 = 888$

$45 \times 100 = 4500$

$123 \times 6 = 738$

7 Dividing

Short division: divide one digit at a time, carrying what's left.

$84 \div 4 = 21$ $936 \div 8 = 117$

$47 \div 5 = 9$ remainder 2

Check: $117 \times 8 = 936$ ✓

8 Estimating and rounding

Look at the next digit: 5 or more rounds up.

387 to the nearest 10 is **390**, to the nearest 100 is **400**

Estimate $48 \times 21 \approx 50 \times 20 = 1000$

Estimate $612 \div 3 \approx 600 \div 3 = 200$

9 Order of operations

Brackets first, then × and ÷ from left to right, then + and − from left to right.

$3 + 4 \times 5 = 23$ $(3 + 4) \times 5 = 35$

$20 - 12 \div 4 = 17$

$2 \times (6 + 4) - 5 = 15$

Watch out: don't just work from left to right.

YEAR 7 MATHS · CHAPTER 2 · MIND MAP

Number properties and patterns

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1 Factors and multiples

Factors go into a number. Multiples come out of its times table.

Factors of 12: 1, 2, 3, 4, 6, 12

Multiples of 4: 4, 8, 12, 16...

2 HCF and LCM

Highest common factor: the biggest number into both. Lowest common multiple: the smallest both go into.

12 and 18: HCF is **6**, LCM is **36**

3 Divisibility tests

By 2: even. By 3: digits add to a multiple of 3. By 5: ends in 0 or 5. By 9: digits add to a multiple of 9.

126: digits add to 9, so it divides by 3 and 9

4 Prime numbers

A prime has exactly two factors: 1 and itself.

2, 3, 5, 7, 11, 13, 17, 19

1 isn't prime. 2 is the only even prime.

5 Using indices

The index counts how many times the base multiplies: $5 \times 5 \times 5 = 5^3$.

$2^4 = 16 \quad 3^3 = 27 \quad 10^3 = 1000$

Watch out: $2^3 = 2 \times 2 \times 2 = 8$, not 6.

6 Prime decomposition

Use a factor tree until every branch ends in a prime.

$36 = 2^2 \times 3^2$

$90 = 2 \times 3^2 \times 5$

7 Squares and square roots

$7^2 = 49$, so $\sqrt{49} = 7$. They undo each other.

$\sqrt{144} = 12$

Perfect squares: 1, 4, 9, 16, 25, 36...

8 Number patterns

Find what happens each step: add a number, or multiply by one.

3, 7, 11, 15: add 4, next is **19**

2, 6, 18: multiply by 3, next is **54**

9 Spatial patterns

Count how much each new shape adds, then write a rule.

Squares from matchsticks: 4, 7, 10, so $\text{sticks} = 3n + 1$

10 squares: $3 \times 10 + 1 = 31$ sticks

10 Tables and rules

The step in the output tells you what to multiply by.

Input 1, 2, 3 and output 5, 8, 11: $y = 3x + 2$

$x = 10$: $y = 32$

11 The Cartesian plane

(x, y): across first, then up. The origin is $(0, 0)$.

$(3, 2)$ is 3 across and 2 up

Plot $y = x + 1$: $(0, 1), (1, 2), (2, 3)$

YEAR 7 MATHS · CHAPTER 3 · MIND MAP

Fractions and percentages

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1 Introduction to fractions

The bottom says how many equal parts. The top says how many you have.

$\frac{3}{4}$: 3 of 4 equal parts

Improper: $\frac{5}{4}$ is more than a whole

2 Equivalent fractions

Multiply or divide the top and bottom by the same number.

$\frac{3}{4} = \frac{6}{8}$

Simplify: $\frac{12}{18} = \frac{2}{3}$

3 Mixed and improper

Multiply the whole by the bottom, then add the top.

$2\frac{1}{3} = \frac{7}{3}$

$\frac{11}{4} = 2\frac{3}{4}$

4 Ordering fractions

Rewrite with a common denominator, then compare the tops.

$\frac{2}{3}$ and $\frac{3}{4}$: $\frac{8}{12} < \frac{9}{12}$, so $\frac{2}{3} < \frac{3}{4}$

5 Adding fractions

Common denominator first, then add the tops only.

$\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$

$\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

$1\frac{1}{2} + 2\frac{1}{3} = 3\frac{5}{6}$

Watch out: don't add the bottoms: $\frac{1}{2} + \frac{1}{3} \neq \frac{2}{5}$

6 Subtracting fractions

Common denominator first, then subtract the tops.

$\frac{3}{4} - \frac{1}{3} = \frac{9}{12} - \frac{4}{12} = \frac{5}{12}$

$2\frac{1}{2} - \frac{3}{4} = \frac{10}{4} - \frac{3}{4} = 1\frac{3}{4}$

7 Multiplying fractions

Top times top, bottom times bottom. No common denominator needed.

$\frac{2}{3} \times \frac{3}{5} = \frac{6}{15} = \frac{2}{5}$

$\frac{1}{2}$ of 12 is 6

8 Dividing fractions

Flip the second fraction, then multiply.

$\frac{3}{4} \div \frac{1}{8} = \frac{3}{4} \times 8 = 6$

$\frac{2}{3} \div 4 = \frac{2}{3} \times \frac{1}{4} = \frac{1}{6}$

9 Fractions and percentages

Per cent means out of 100.

$\frac{1}{4} = 25\%$ $\frac{3}{5} = 60\%$

$35\% = \frac{35}{100} = \frac{7}{20}$

10 Percentage of a number

Find 10% first, or multiply by the percentage as a fraction.

10% of 60 is 6, so 20% of 60 is **12**

15% of 80: $\frac{15}{100} \times 80 = 12$

11 Introduction to ratios

A ratio compares amounts. Order matters.

3 red and 5 blue: red to blue is **3 : 5**

Simplify: $6 : 8 = 3 : 4$

12 Problems with ratios

Add the parts, find one part, then multiply.

$40 in the ratio 3 : 5: 8 parts, 1 part = $5

So **$15** and **$25**

YEAR 7 MATHS · CHAPTER 4 · MIND MAP

Algebraic techniques

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1 Introduction to algebra

A pronumeral is a letter for a number. $3x$ means $3 \times x$.

"7 more than n": $n + 7$

"Double y": $2y$

"x divided by 4": $\frac{x}{4}$

In $3x + 5$: the coefficient is 3, the constant is 5

2 Substituting

Replace each letter with its number, then work it out.

$x = 4$: $3x + 2 = 14$ $x^2 = 16$

$20 \div x = 5$

$a = 2,\ b = 5$: $ab + 1 = 11$

Watch out: $3x$ with $x = 4$ is 3×4, not 34.

3 Equivalent expressions

Equivalent expressions are equal for every value of the letter.

$2(x + 3)$ and $2x + 6$

Test with $x = 5$: both give 16 ✓

$x + x = 2x$

$3a + 2a = 5a$

4 Like terms

Only terms with the same letters and powers can be added or subtracted.

$3x + 2x = 5x$

$4a + 2b - a = 3a + 2b$

$5x$ and x^2 aren't like terms

$2x + 3 + x + 4 = 3x + 7$

5 Multiplying and dividing

Multiply the numbers, then the letters. Divide like a fraction.

$3 \times 2x = 6x$ $2a \times 3b = 6ab$

$x \times x = x^2$

$8x \div 2 = 4x$ $12ab \div 4a = 3b$

6 Expanding brackets

Multiply everything inside by the term outside.

$3(x + 2) = 3x + 6$

$5(2a - 1) = 10a - 5$

$x(x + 4) = x^2 + 4x$

Watch out: multiply the second term too.

7 Applying algebra

Pick a letter for the unknown and write a rule.

Gym: \$10 to join plus \$5 a visit: $C = 10 + 5v$. 8 visits: \$50

Rectangle, width x, length $x + 2$: perimeter $4x + 4$. If $x = 3$, that's 16.

30 students on each of b buses: $30b$. 4 buses: 120 students

YEAR 7 MATHS · CHAPTER 5 · MIND MAP

Decimals

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1 Place value and decimals

After the point: tenths, hundredths, thousandths.

3.47 is 3 ones, 4 tenths and 7 hundredths

0.5 is bigger than 0.45

Watch out: more digits doesn't mean bigger.

2 Rounding decimals

Look at the next digit: 5 or more rounds up.

3.476 to 1 decimal place is **3.5**

6.849 to 2 decimal places is **6.85**

2.93 to the nearest whole is **3**

3 Adding and subtracting

Line up the decimal points. Fill gaps with zeros.

$3.5 + 2.75 = 6.25$

$5 - 1.8 = 3.2$

4 × and ÷ by 10, 100, 1000

Multiply: digits move left. Divide: digits move right. One place for each zero.

$3.4 \times 100 = 340$

$56 \div 10 = 5.6$

$0.7 \div 100 = 0.007$

5 Multiplying decimals

Multiply as whole numbers, then count the decimal places.

$0.3 \times 0.4 = 0.12$ **(two places)**

$1.2 \times 3 = 3.6$

$2.5 \times 1.2 = 3$

6 Dividing decimals

Make the number you're dividing by whole: move both points the same way.

$7.5 \div 3 = 2.5$

$4.2 \div 0.7 = 42 \div 7 = 6$

7 Decimals and fractions

To turn a fraction into a decimal, divide the top by the bottom.

$0.75 = \frac{3}{4}$ $0.2 = \frac{1}{5}$

$\frac{3}{8} = 0.375$ $\frac{2}{3} = 0.666\ldots$

8 Decimals and percentages

Decimal to percentage: multiply by 100. Percentage to decimal: divide by 100.

$0.35 = 35\%$ $1.2 = 120\%$

$7\% = 0.07$

9 Expressing proportions

Write the part over the whole, then convert.

12 out of 20: $\frac{12}{20} = \frac{3}{5} = 0.6 = 60\%$

9 out of 25: $\frac{9}{25} = 0.36 = 36\%$

YEAR 7 MATHS · CHAPTER 6 · MIND MAP

Negative numbers

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1 Introduction to negative integers

Numbers get smaller as you move left on the number line.

$-3 < 1 \quad -7 < -2$

4 degrees below zero is −4°C

In order: −8, −3, 0, 2

The opposite of 5 is −5

A lift at floor 2 goes down 5 floors: floor **−3**

2 Adding and subtracting positives

Adding moves right. Subtracting moves left.

$-3 + 5 = 2 \quad -2 + 6 = 4$

$2 - 7 = -5 \quad -4 - 3 = -7$

It's −3°C and warms by 8°: **5°C**

A diver at −12 m rises 5 m: **−7 m**

3 Adding and subtracting negatives

Adding a negative means subtract. Subtracting a negative means add.

$3 + (-5) = -2 \quad -1 + (-4) = -5$

$2 - (-3) = 5 \quad -6 - (-2) = -4$

Owing $5, then owing $4 more: $-5 + (-4) = -9$

Watch out: $+(-)$ acts like $-$, and $-(-)$ acts like $+$.

4 Multiplying and dividing

Same signs give a positive. Different signs give a negative.

$-3 \times 4 = -12 \quad -5 \times (-2) = 10$

$-20 \div 4 = -5 \quad -18 \div (-3) = 6$

Lose 3 points in each of 4 rounds: $-3 \times 4 = -12$

$-2 \times (-3) \times 4 = 24$

Watch out: this rule is for × and ÷ only: $-3 + (-2) = -5$.

5 Order of operations

Brackets, then × and ÷, then + and −, from left to right.

$-2 + 3 \times (-4) = -2 - 12 = -14$

$(-2 + 3) \times (-4) = -4$

$10 \div (-2) - 3 = -8$

$-3 \times 2 + 8 = -6 + 8 = 2$

Do the multiplying and dividing before the adding and subtracting.

6 Substituting integers

Put negative numbers in brackets when you substitute.

$a = -2,\ b = 3$: $a + b = 1 \quad ab = -6$

$b - a = 3 - (-2) = 5 \quad a^2 = 4$

$2a - b = -4 - 3 = -7$

$ab + 10 = -6 + 10 = 4$

Watch out: $(-2)^2 = 4$, not -4.

7 The Cartesian plane

Four quadrants, numbered anticlockwise from the top right. (x, y): across, then up or down.

$(-3, 2)$ is in the second quadrant

$(-1, -4)$ is in the third quadrant

$(2, -3)$ is in the fourth quadrant

Right 5 from $(-3, 2)$ gives $(2, 2)$

$(-2, 0)$ is on the x-axis, so it isn't in any quadrant.

YEAR 7 MATHS · CHAPTER 7 · MIND MAP

Geometry

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1 Points, lines and angles

Acute under 90°, right 90°, obtuse 90° to 180°, straight 180°, reflex over 180°.

A full revolution is 360°

2 Adjacent and opposite angles

On a straight line, angles add to 180°. Vertically opposite angles are equal.

130° on a straight line: the other angle is **50°**

Complementary angles add to 90°

3 Parallel lines

Corresponding angles are equal (F), alternate angles are equal (Z), co-interior angles add to 180° (C).

75°: co-interior angle **105°**

4 Compound problems

Draw an extra parallel line through the angle to split it.

Parts of 40° and 30°: the whole angle is **70°**

5 Triangles

By sides: equilateral, isosceles, scalene. By angles: acute, right, obtuse.

Equilateral: three equal sides

Isosceles: two equal sides

Scalene: no equal sides

6 Quadrilaterals and polygons

Square, rectangle, parallelogram, rhombus, trapezium, kite.

Polygons: pentagon 5 sides, hexagon 6, octagon 8

Rhombus: four equal sides

7 Angle sum of a triangle

The angles of a triangle add to 180°.

50° and 60°: the third is **70°**

Isosceles with top angle 40°: each base angle is **70°**

8 Symmetry

Line symmetry: it folds onto itself. Rotational symmetry: it fits onto itself as it turns.

Square: 4 lines, order 4

Rectangle: 2 lines, order 2

9 Reflection and rotation

Reflection: a mirror image. Rotation: a turn about a centre.

Give a rotation's centre, angle and direction, such as 90° clockwise

10 Translation

A slide, with no turning or flipping.

Right 4, up 1: $(2, 1) \rightarrow (6, 2)$

11 Drawing solids

Prisms, pyramids, cylinders, cones and spheres.

Draw top, front and side views

Isometric dot paper makes 3D drawing easier

12 Nets of solids

A net folds up into the solid.

Cube: 6 squares

Triangular prism: 2 triangles and 3 rectangles

Square pyramid: 1 square and 4 triangles

YEAR 7 MATHS · CHAPTER 8 · MIND MAP

Statistics and probability

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1 Collecting and classifying data

Categorical data is groups. Numerical data is numbers.

Categorical: favourite sport

Numerical: discrete (counts), continuous (measurements)

Primary: you collect it. Secondary: someone else did.

2 Summarising data

Mean: add and divide. Median: middle in order. Mode: most common. Range: max – min.

2, 3, 3, 6, 11: mean $\frac{25}{5} = 5$

Median 3, mode 3, range 9

3 Column graphs and dot plots

Give it a title, label both axes and use an equal scale.

Column graphs: leave gaps between columns

Dot plots: one dot for each result

4 Line graphs

Show change over time, with time on the horizontal axis.

Look for rising, falling and flat parts

5 Stem-and-leaf plots

The stem is the tens, the leaf is the units. Include a key.

1 | 2 5 8 and 2 | 0 3 (key 1 | 2 = 12)

Data 12, 15, 18, 20, 23: median **18**

6 Sector and divided bar graphs

Each part of the circle or bar shows a share of the whole.

A quarter of a circle is $90°$

30% of a circle: $0.3 \times 360° = 108°$

7 Describing probability

Probability runs from 0 (impossible) to 1 (certain).

Unlikely, even chance $\left(\frac{1}{2}\right)$, likely

8 Theoretical probability

$P = \frac{\text{favourable}}{\text{total}}$ when outcomes are equally likely.

Fair die: $P(3) = \frac{1}{6}$, $P(\text{even}) = \frac{1}{2}$

3 red and 2 blue: $P(\text{red}) = \frac{3}{5}$

$P(\text{not } A) = 1 - P(A)$

9 Experimental probability

Relative frequency $= \frac{\text{times it happened}}{\text{number of trials}}$

12 heads in 40 tosses: $\frac{12}{40} = 0.3$. The theoretical probability is 0.5.

More trials usually get closer to the theoretical probability.

YEAR 7 MATHS · CHAPTER 9 · MIND MAP

Equations

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1 Introduction to equations

An equation says two things are equal. The solution makes it true.

$x + 3 = 7$ is true when $x = 4$

$2x = 10$ is true when $x = 5$

Is $x = 3$ a solution of $2x + 1 = 7$?
$2(3) + 1 = 7$ ✓ yes

2 Solving by inspection

Ask: what number makes this true?

$x + 5 = 12 \Rightarrow x = 7$

$3x = 18 \Rightarrow x = 6$

$\frac{x}{2} = 4 \Rightarrow x = 8$

$x - 4 = 9 \Rightarrow x = 13$

3 Equivalent equations

Do the same to both sides and the equation stays balanced.

$x + 3 = 7$: subtract 3, $x = 4$

$2x = 8$: divide by 2, $x = 4$

$x - 5 = 2$: add 5, $x = 7$

$\frac{x}{3} = 2$: multiply by 3, $x = 6$

4 Solving algebraically

Undo each operation in reverse order, then check by substituting.

$2x + 3 = 11 \Rightarrow 2x = 8 \Rightarrow x = 4$

$5x - 2 = 13 \Rightarrow 5x = 15 \Rightarrow x = 3$

$4x + 1 = 21 \Rightarrow 4x = 20 \Rightarrow x = 5$

Check: $2(4) + 3 = 11$ ✓

5 Equations with fractions

Multiply both sides to clear the fraction.

$\frac{x}{3} = 4 \Rightarrow x = 12$

$\frac{x}{2} + 1 = 5 \Rightarrow \frac{x}{2} = 4 \Rightarrow x = 8$

$\frac{x+2}{3} = 4 \Rightarrow x + 2 = 12 \Rightarrow x = 10$

6 Equations with brackets

Expand the brackets first.

$2(x + 3) = 14 \Rightarrow 2x + 6 = 14 \Rightarrow x = 4$

$3(x - 1) = 12 \Rightarrow 3x - 3 = 12 \Rightarrow x = 5$

$4(x + 2) = 20 \Rightarrow 4x + 8 = 20 \Rightarrow x = 3$

7 Using formulas

Substitute what you know, then solve for what's left.

$A = lw$, $l = 6,\ w = 4$: $A = 24$

$P = 2l + 2w$, $P = 20,\ l = 6$: $20 = 12 + 2w$, so $w = 4$

$D = ST$, $S = 60,\ T = 2$: $D = 120$

8 Solving problems

Use a letter for the unknown, write an equation, solve it, answer in words.

A number plus 9 is 21: $x + 9 = 21$, so the number is 12

3 pens plus a $5 ruler cost $14: $3x + 5 = 14$, so each pen is $\$3$

Rectangle, width w, length $w + 3$, perimeter 26: $4w + 6 = 26$, so $w = 5$: 5 by 8

YEAR 7 MATHS · CHAPTER 10 · MIND MAP

Measurement

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1 Metric units of length

1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm.

2.5 m = **250 cm**

3400 m = **3.4 km**

45 mm = **4.5 cm**

Big unit to small: multiply. Small unit to big: divide.

2 Perimeter

Add all the sides.

Rectangle 7 by 4: $2(7+4) = 22$

Square side 5: $4 \times 5 = 20$

Regular hexagon, side 3: $6 \times 3 = 18$

3 Circles and circumference

$C = \pi d = 2\pi r$. The radius is half the diameter.

$d = 6$: $C = 6\pi \approx 18.8$

$r = 4$: $C = 8\pi \approx 25.1$

$\pi \approx 3.14$

4 Arcs and sectors

Arc length $= \frac{\theta}{360} \times 2\pi r$. Add the straight edges for the perimeter.

Semicircle $r = 5$: arc $5\pi \approx 15.7$, perimeter $10 + 5\pi \approx 25.7$

Quarter circle $r = 4$: arc $2\pi \approx 6.3$, perimeter $8 + 2\pi \approx 14.3$

5 Area of rectangles

$A = lw$, in square units.

$8 \times 3 = 24$ cm²

1 cm² = 100 mm², 1 m² = 10 000 cm²

Square, side 5: 25 cm²

2 m by 3 m: 6 m²

Watch out: to convert area units, square the length conversion.

6 Area of parallelograms

$A = bh$, with the height at right angles to the base.

Base 6, height 4: $A = 24$

Base 8 cm, height 5 cm: $A = 40$ cm²

Watch out: use the perpendicular height, not the slanted side.

7 Area of triangles

$A = \frac{1}{2}bh$: half a parallelogram.

Base 10, height 3: $A = 15$

Base 6, height 7: $A = 21$

Watch out: don't forget the $\frac{1}{2}$.

8 Composite shapes

Split into shapes you know and add, or take a shape away.

L shape: $8 \times 3 + 4 \times 2 = 24 + 8 = 32$

10 by 6 with a 2 by 2 hole: $60 - 4 = 56$

9 Volume of rectangular prisms

$V = lwh$, in cubic units.

$5 \times 3 \times 2 = 30$ cm³

Cube, side 3: $3 \times 3 \times 3 = 27$ cm³

10 Volume of triangular prisms

$V = Ah$: area of the triangle times the length.

Triangle $\frac{1}{2} \times 4 \times 3 = 6$, length 10: $V = 60$

Triangle base 6, height 5: area 15. Length 4: $V = 60$

11 Capacity

1 L = 1000 mL, 1 cm³ = 1 mL, 1 m³ = 1000 L.

250 mL = **0.25 L**

A 10 cm cube holds 1000 cm³, or **1 L**

1.5 L = **1500 mL**

12 Mass and temperature

1 t = 1000 kg, 1 kg = 1000 g.

2.5 kg = **2500 g**

Water freezes at 0°C and boils at 100°C

750 g = **0.75 kg**

From $-2°$C to $5°$C is a rise of $7°$